Linux System Administration

Exam Quick Revision & Practical Command Guide

CSE 493 · Units I–VI

Prepared from the course syllabus, Linux presentations, practical questions, shell notes, networking/storage examples, and revision files in **linuxppt1**.

How to use tonight: Read the command tables once, practise the 15 Unit I–II scenarios, then revise the bold “must remember” points. Never run destructive storage commands on a real machine without confirming the device.

Linux commands may vary slightly between RHEL versions. Use `man command` and `command --help` when in doubt.

1. Exam strategy and command pattern

Scenario → command → verify. For almost every practical, first inspect the current state, make one change, then verify with a read-only command. Use `sudo` only when required.

Need	Useful commands
Current location and files	`pwd`, `ls -la`, `cd /path`
Help	`man cmd`, `cmd --help`, `info cmd`, `apropos keyword`
Verify identity	`whoami`, `id`, `hostname`
Read safely	`cat`, `less`, `head`, `tail`
Find and filter	`find`, `grep`, `grep -E`, pipes `|`

2. Unit I — command line, files, help, users

Navigation and file operations

```
pwd

cd /etc

ls -lah

cd /var/log

cd ~                    # home directory

mkdir -p project/{commands,reports,backup}
```

```
touch project/commands/{cmd1.txt,cmd2.txt,cmd3.txt}

cp source.txt backup/

mv old.txt new.txt

rm -i file.txt       # ask before deleting

rm -r directory      # recursive: use carefully
```

Finding files and shell expansion

```
find /etc -type f -name '*.conf' 2>/dev/null

find . -type f -iname '*report*'

ls data*.txt

ls report?.txt

# redirection

printf 'pwd\nls\nid\n' > linux_commands.txt

printf 'date\nwhoami\n' >> linux_commands.txt

cat linux_commands.txt
```

Text and documentation

```
less file.txt

head -n 10 file.txt

tail -f /var/log/messages

man chmod                 # search inside man: /pattern; quit: q

man -k password           # keyword search (same idea as apropos)

apropos permissions

info coreutils 'ls'
```

Users and groups

```
sudo groupadd developers

sudo useradd -m developer1

sudo useradd -m developer2

sudo usermod -aG developers developer1

sudo usermod -aG developers developer2

sudo passwd developer1

id developer1

groups developer1

getent group developers
```

```
sudo userdel -r developer1   # only when deletion is intended
```

Remember: `usermod -aG` means append supplementary groups. Without `-a`, existing supplementary groups can be replaced.

3. Unit II — permissions, processes, services, SSH, logs

Permissions and ownership

```
ls -l project.txt

chmod 640 project.txt       # owner rw, group r, others none

chmod 750 shared_lab        # owner rwx, group r-x, others none

chmod u+rw,g+r,o-rwx file

sudo chown user:group file

umask

# numeric: r=4, w=2, x=1; 640 = rw-r-----
```

Object	Read	Write	Execute
File	view contents	change contents	run as program
Directory	list names	create/delete entries	enter/traverse

Processes and jobs

```
ps -ef

ps -u "$USER"

top

pgrep -a process_name

sleep 300                 # Ctrl+Z suspends

jobs

bg %1

fg %1

kill PID                  # graceful TERM

kill -9 PID               # force only if necessary

systemctl status sshd
```

Services and SSH

```
sudo systemctl enable --now sshd
```

```
systemctl is-enabled sshd

systemctl is-active sshd

ssh user@localhost

whoami; hostname; pwd

exit

ssh-keygen -t ed25519

ssh-copy-id user@server

ssh user@server
```

Service name: RHEL commonly uses `sshd`. If unsure, use `systemctl list-unit-files | grep -i ssh`.

Logs and time

```
journalctl -n 50

journalctl -u sshd

journalctl -u sshd -r          # newest first

journalctl -p warning -b       # warnings/errors this boot

journalctl --since '1 hour ago'

systemctl status systemd-journald

timedatectl
```

4. Unit III — networking, archives, packages, filesystems

Network inspection and nmcli

```
nmcli device status

nmcli connection show

ip addr

ip route

ping -c 4 8.8.8.8

sudo nmcli con add con-name testing type ethernet ifname ens160

sudo nmcli con mod testing ipv4.method manual

sudo nmcli con mod testing ipv4.addresses 192.168.120.150/24

sudo nmcli con mod testing ipv4.gateway 192.168.120.2

sudo nmcli con mod testing ipv4.dns 8.8.8.8

sudo nmcli con up testing
```

Archives, packages, links

```
tar -czf backup.tar.gz directory/

tar -tzf backup.tar.gz

tar -xzf backup.tar.gz

scp file user@host:/tmp/

rsync -av directory/ user@host:/backup/

sudo dnf search keyword

sudo dnf install package

sudo dnf update

rpm -q package

rpm -ql package

ln -s target linkname

readlink linkname
```

Mounting filesystems

```
lsblk

sudo fdisk -l

sudo blkid

sudo mkdir -p /data

sudo mkfs.xfs /dev/sdb1       # destroys data on that partition

sudo mount /dev/sdb1 /data

df -hT

findmnt /data

sudo umount /data

# persistent mount: use UUID from blkid in /etc/fstab

UUID=XXXX  /data  xfs  defaults  0 0

sudo mount -a
```

Safety: Confirm the device with `lsblk` before `fdisk`, `mkfs`, `mount`, `umount`, or `mkswap`. A wrong device can erase data.

5. Unit IV — Bash, loops, regex, scheduling, performance, ACL

Bash script template

```
#!/bin/bash
```

```
read -p 'Marks: ' marks

if [ "$marks" -ge 40 ]; then

  echo 'PASS'

else

  echo 'FAIL'

fi

# save as check.sh, then:

chmod +x check.sh

./check.sh
```

Loops and conditions

```
for n in 1 2 3; do echo "$n"; done

for f in *.txt; do echo "$f"; done

n=1

while [ "$n" -le 5 ]; do echo "$n"; n=$((n+1)); done

if [ -f file.txt ]; then echo exists; fi

[ "$a" -gt "$b" ] && echo greater
```

Regular expressions and grep

```
grep '^ERROR' app.log       # beginning of line

grep 'failed$' app.log      # end of line

grep -E 'error|failed' app.log

grep -E '^[0-9]{5}$' file.txt

grep -i 'warning' log.txt

grep -v '^#' config.txt       # exclude comments
```

Regex	Meaning
`^`, `$`	start, end of line
`.`	any one character
`[abc]`, `[0-9]`	one character from a set/range
`*`, `+`, `?`	zero-or-more, one-or-more, optional (extended regex)
`{n}`, `{n,m}`	exact/ranged repetition with `grep -E`

Scheduling and ACL

```
at now + 5 minutes

# type commands, then Ctrl+D

atq

at -c JOB_ID

atrm JOB_ID

crontab -e

# minute hour day month weekday command

setfacl -m u:alice:rw file

getfacl file

setfacl -m g:developers:rX directory

setfacl -b file
```

6. Unit V — SELinux, partitions, swap, LVM, Stratis/VDO

SELinux quick checks

```
getenforce

sestatus

sudo setenforce 0       # temporary permissive; use only when instructed

ls -Z

sudo restorecon -Rv /var/www/html

getsebool -a

sudo ausearch -m AVC -ts recent
```

Partitions, swap, and LVM concepts

```
lsblk

sudo fdisk /dev/sdb

sudo mkfs.xfs /dev/sdb1

sudo mkswap /dev/sdb2

sudo swapon /dev/sdb2

swapon --show

sudo swapoff /dev/sdb2

# LVM conceptual workflow:

sudo pvcreate /dev/sdb1
```

```
sudo vgcreate vgdata /dev/sdb1

sudo lvcreate -n lvdata -L 2G vgdata

sudo mkfs.xfs /dev/vgdata/lvdata

sudo mount /dev/vgdata/lvdata /data

sudo lvextend -r -L +1G /dev/vgdata/lvdata
```

Persistent configuration: use UUIDs in `/etc/fstab`; test with `mount -a`. For swap, the fstab entry uses `none swap defaults 0 0`.

7. Unit VI — NFS, boot, firewall, installation

```
# NFS client examples

showmount -e nfs-server

sudo mkdir -p /mnt/share

sudo mount -t nfs nfs-server:/export /mnt/share

findmnt /mnt/share

# automount is normally configured with autofs and /etc/auto.master


# boot and troubleshooting

systemctl get-default

sudo systemctl set-default multi-user.target

journalctl -b

# firewall

sudo firewall-cmd --state

sudo firewall-cmd --list-all

sudo firewall-cmd --permanent --add-service=http

sudo firewall-cmd --reload

# SELinux port labels

semanage port -l | grep ssh

sudo semanage port -a -t ssh_port_t -p tcp 2222
```

8. Fifteen practical scenarios — model answers

1. `pwd; cd /etc; ls -l; cd /var/log; cd ~`
2. `find /etc -type f -name '*.conf' 2>/dev/null`
3. `mkdir -p ~/LinuxAssessment/{commands,reports,backup}; touch`

`~/LinuxAssessment/commands/cmd{1,2,3}.txt; find ~/LinuxAssessment -print`

4. `touch data{1,2,3}.txt report{1,2}.txt; printf '%s\n' data*.txt; printf '%s\n' report*.txt`
5. `printf 'pwd\nls\nid\n' > linux_commands.txt; cat linux_commands.txt; printf 'date\nwhoami\n' >> linux_commands.txt`
6. `man chmod`; search inside with `/OPTIONS`; installed keyword search with `apropos keyword` or `man -k keyword`.
7. `sudo groupadd developers; sudo useradd -m developer1; sudo useradd -m developer2; sudo usermod -aG developers developer1 developer2; getent group developers`
8. `sudo useradd -m labuser; sudo passwd labuser; id labuser; getent passwd labuser`
9. `touch project.txt; chmod 640 project.txt; stat -c '%A %a %n' project.txt`
10. `mkdir shared_lab; chmod 750 shared_lab; stat -c '%A %a %n' shared_lab`
11. Run `sleep 300`; press `Ctrl+Z`; run `bg; jobs; fg`.
12. `ps -u "$USER"; kill PID; ps -p PID`
13. `systemctl status sshd; sudo systemctl enable sshd; systemctl is-enabled sshd`
14. `ssh "$USER"@localhost; whoami; hostname; pwd; exit`
15. `journalctl -n 50; journalctl -u sshd; journalctl -u sshd -r; journalctl -p warning -b`

9. Last-minute checklist

- Can you explain the permission digits 640 and 750?
- Can you distinguish `find` from `grep`?
- Can you use `systemctl` to inspect, start, stop, enable, and verify a service?
- Can you inspect a network using `nmcli`, `ip`, and `ping`?
- Can you create a filesystem, mount it, and make it persistent using UUID/fstab?
- Can you explain SELinux enforcing/permissive and read an AVC denial?
- Can you write an `if` script and a `for`/`while` loop?
- Can you schedule one deferred job with `at` and recurring jobs with `cron`?
- For every command, can you state how you will verify the result?

End of revision guide · Use the original presentations and syllabus for deeper explanations.